

*The voice that uses your computer for you*

"yes?"



Ctrl+Alt+M



A voice agent for Windows, heard by *AssemblyAI*, for people who can't see the screen.  
team revived · lablab.ai × AssemblyAI Voice Agent Hackathon · September 2026

# “Button, button, *button.*”

That is what a screen reader says when an app doesn't label its buttons. Many don't. Blind users then need a sighted person to do it for them, and older users give up.

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## ~95%

of the top million home pages fail  
basic accessibility checks

Source: WebAIM Million, 2024-25.



business value

An app that wasn't  
built for you *costs 260  
hours a year.*

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€4,000

and 260 hours a year per  
person, in sighted help and  
lost time (our estimate)

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338 M

people blind or with  
moderate to severe vision  
loss

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\$411 bn

productivity lost to vision  
impairment every year

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1 in 6

people will be over 60 by  
2030, the second audience

Estimate: 4 h a week of sighted help + 1 h lost, at €15/h, plus missed online prices. Sources: Bourne et al., Lancet Global Health 2021, GBD 2019 (43 M blind + 295 M); Marques et al., eClinicalMedicine 2021; WHO.

*business value*

# Say it. *Iris does it*, and asks first.

You say what you want in plain words. Iris does it in any app on your PC, tells you what happened, and waits for your spoken yes before anything that sends, pays or deletes.

|      |                                                            |
|------|------------------------------------------------------------|
| YOU  | Put vitamin D in my basket.                                |
| IRIS | Done. One bottle of vitamin D3 is in your basket.          |
| YOU  | Now check out.                                             |
| IRIS | <i>Check out and pay for your basket? It costs €32.86.</i> |
| YOU  | No, wait.                                                  |
| IRIS | Okay. I didn't buy anything.                               |

# Screen readers wait for developers. *Iris doesn't.*

| LAYER            | USED WHEN                                                                             | NEEDS THE APP'S DEVELOPER? |
|------------------|---------------------------------------------------------------------------------------|----------------------------|
| 1 · An API       | The service has one: Gmail through MCP, web pages through browser shortcuts           | No                         |
| 2 · The tree     | Jev (or Kev on your own GPU) picks from the controls Windows already sees             | No                         |
| 3 · The keyboard | Enter sends, Escape closes, Alt+Left goes back                                        | No                         |
| 4 · The pixels   | With Claude: unlabelled icons and custom-drawn apps, seen like a sighted helper would | No                         |

Proof: Telegram, Amazon, Gmail. **10 of 10** unlabelled icons found with Claude vision.

*originality*

# Three errands, *real services*.

| ERRAND   | WHAT YOU HEAR                                                                                                            | WHAT IT PROVES                  |
|----------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Gmail    | “Did Peter write?” It reads the email, writes your reply, asks “Send it to Peter?”, and asks again when you sound unsure | API first, spoken yes           |
| Telegram | “Tell Peter I'm running late.” The send button has no label; Iris still sends it                                         | Works where screen readers stop |
| Amazon   | Fills the basket, names the price, and “No, wait” means nothing is bought                                                | Money needs a clear yes         |

Every clip is a real, continuous take on test accounts; sped-up clips are marked 2x. Video and live page: [whom-m0rty.github.io/Iris-voice-agent](https://whom-m0rty.github.io/Iris-voice-agent)

*originality*

# Both AssemblyAI paths, *one agent*.

## REALTIME STT • THE DEMO

### Universal-Streaming

- Speech-to-text over WebSocket, formatted turns
- Pauses merged into one turn: older people pause mid-sentence (0.8–4 s, set by voice)
- Speech-start events let you talk over Iris
- Our own brain (Qwen, or Claude) and neural TTS

## VOICE AGENT API

### One connection

- Speech, turn-taking and voice in one WebSocket
- JSON-Schema tools: do\_task, answer\_confirmation, stop\_task, mute\_microphone
- Tools answer at once; progress spoken with reply.create
- Runs as a second backend

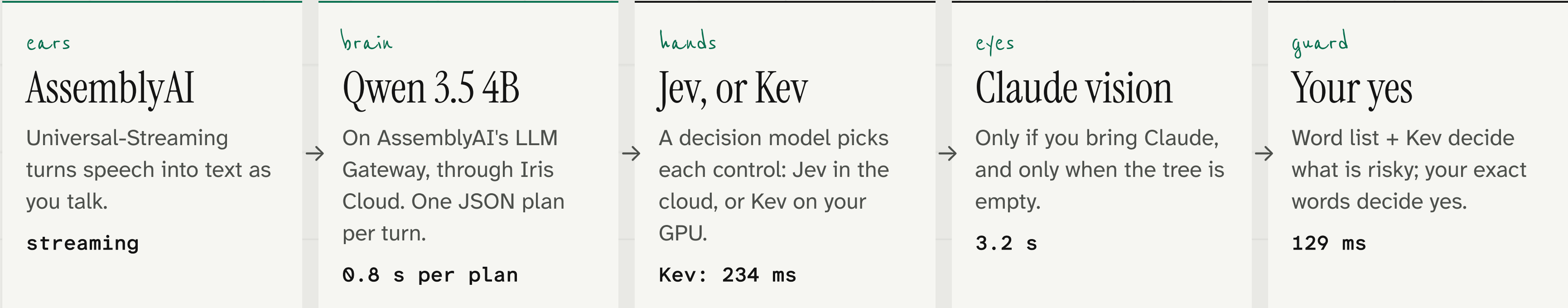
## LLM GATEWAY

### The brain

- Qwen 3.5 4B by default: about 0.8 s per plan
- Through Iris Cloud, our server on Oracle Ampere: it holds the keys, so your PC needs none
- Any of the 47 gateway models, or Claude, with one line in .env

*application of technology*

# Each model does *what it's best at.*



Iris Cloud, our small server on Oracle Ampere, holds the keys: no keys and no GPU on your PC. Audio goes straight to AssemblyAI. Latencies: Qwen from the end-to-end runs (27.09); Kev, vision and yes/no from bench/bench\_full.py.

*application of technology*

# It never acts *on a maybe.*

---

0

false yeses in 60 spoken answers

---

0

risky actions let through, in 40 it had never seen

---

92%

controls picked right by Kev in 25 test steps; misses went to vision, not to a wrong click

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10/10

unlabelled icons found by Claude vision

Passwords never reach a model: code types them from Windows Credential Manager. Apps on the do-not-touch list are never read or clicked. Muted, a pending question can only end as a no.

# Small behaviours *that change the day.*

*“What’s on my screen?”*

Where you are, pop-ups first, three things that matter. Photos described, with Claude.

*“I’m lost.”*

Says where you are and what box is open. Never closes it without asking.

*“Show me how.”*

Does the task slowly, explaining each step in plain words. It teaches as it goes.

*“Say that again.”*

Repeats the last reply word for word, at once.

*“Wait longer for me.”*

Pause length set by voice, remembered next time.

*tick · chime · low tone*

Sounds for thinking, done and error. Silence never leaves you guessing.

# Free for the person. *Paid by who carries the cost today.*



## 1 Families

“Mum calls me every day to fix her email.” Remote setup, a daily log, alerts when she's stuck.

€9-12 / month

## 2 Assistive-tech funders

Rehab agencies, Access to Work, blind associations: budgets exist for approved tools.

€150-400 / seat / year

## 3 Employers

Internal apps that will never be made accessible become usable, with audit log and do-not-touch list.

€20-40 / seat / month

## 4 Banks and telcos

Older customers call support for what the app could do. Connect an API; every call avoided is saved cost.

per active user or task

*business value*

# Law, age and cost *all point one way.*

**The law.** The European Accessibility Act has applied since 28 June 2025 to banking, shopping and telecom.

**Age.** 1 in 6 people over 60 by 2030 (WHO).

**Cost.** Voice AI is now fast and cheap enough to do the task itself, where the alternative is a sighted helper's hour.

| WHEN | MOVE                                                      | PROOF                   |
|------|-----------------------------------------------------------|-------------------------|
| Q1   | Free beta with two blind associations and 20 families     | Tasks done without help |
| Q2   | Family plan, case studies                                 | Month-two retention     |
| Q3   | One funder's approved list; first employer pilot          | First funded seats      |
| Q4   | Bank or telco pilot: bills and balances through their API | Support calls avoided   |

*business value*

# A computer is easy when it *listens* *to you.*

[whom-m0rty.github.io/Iris-voice-agent](https://whom-m0rty.github.io/Iris-voice-agent) · [github.com/Whom-m0rty/Iris-voice-agent](https://github.com/Whom-m0rty/Iris-voice-agent)

